

C1M3_Programming

85 / 100 points earned

24 / 27 autograded cells passed

Graded Cells

Cell 3 (cell-6e8732a9d9253461)

Passed | 2.35 / 2.35 points

View feedback

Cell 5 (cell-d6e7049c70e293d0)

Passed | 2.35 / 2.35 points

View feedback

Cell 7 (cell-356519bae70ac76a)

Passed | 0 / 0 points

View feedback

Cell 9 (cell-904f7f771000969)

Passed | 3.53 / 3.53 points

View feedback

Cell 12 (cell-c8b4d326f2014f9)

Passed | 4.71 / 4.71 points

View feedback

Cell 14 (cell-4e3a9c0c167da76f)

Passed | 2.35 / 2.35 points

View feedback

Cell 16 (cell-1733648e39b1e461)

Passed | 3.53 / 3.53 points

View feedback

Cell 18 (cell-463c0988ae5c35b)

Passed | 2.35 / 2.35 points

View feedback

Cell 21 (cell-32d10be0d080e97c1)

Passed | 5.88 / 5.88 points

View feedback

Cell 23 (cell-abc170c3f4748bbb)

Passed | 2.35 / 2.35 points

View feedback

Cell 25 (cell-51e047320bae4379)

Passed | 2.35 / 2.35 points

View feedback

Cell 27 (cell-6f642d972bcc72a)

Passed | 3.53 / 3.53 points

View feedback

Cell 29 (cell-4302551bbcb36621)

Passed | 3.53 / 3.53 points

View feedback

Cell 31 (cell-2f8a5fb451559a03)

Passed | 2.35 / 2.35 points

View feedback

Cell 33 (cell-31a7e1ef597e0dd3)

Passed | 2.35 / 2.35 points

View feedback

Cell 35 (cell-d6dc58c8f9f91ed)

Passed | 5.88 / 5.88 points

View feedback

Cell 37 (cell-267a26302ebddbba)

Passed | 2.35 / 2.35 points

View feedback

Cell 39 (cell-c562bfcd33da49ea)

Passed | 2.35 / 2.35 points

View feedback

Cell 41 (cell-3fe317a0b4c0c09f)

Passed | 2.35 / 2.35 points

View feedback

Cell 43 (cell-0ec58b70bed70ab5)

Passed | 3.53 / 3.53 points

View feedback

Cell 45 (cell-4d915689d4c8b04e)

Try again | 0.00 / 3.53 points

View feedback

Cell 47 (cell-37b031ea5fb0fb49)

Passed | 2.35 / 2.35 points

View feedback

Cell 49 (cell-f60c37ee3a0438ae)

Try again | 0.00 / 2.35 points

View feedback

Cell 51 (cell-fecfaa82c8ad93fd)

Passed | 9.41 / 9.41 points

View feedback

Cell 53 (cell-ba8effe02f89c36f)

Try again | 0.00 / 9.41 points

View feedback

Cell 55 (cell-5a5b019640dc3734)

Passed | 5.88 / 5.88 points

View feedback

Cell 57 (cell-0ba25422b796451f)

Passed | 7.05 / 7.05 points

View feedback

2. FALSE. Given that the upper end of the confidence interval is almost 30, we can't say we have evidence to suggest $\mu = 30$.

If you believe this statement is true, code `response=TRUE` into the code block below. If you believe this statement to be false, code `response=FALSE` in the code block below. Be sure to use all CAPS after the = sign.

In [32]: Student's answer (Top)

```
response = TRUE
# your code here
# Not yet implemented()
```

In [33]: Grade cell: c11-31a7e1ef597e0dd3 Score: 2.78 / 2.78 (Top)

Hidden Tests Redacted

Congratulations! All test cases in this cell passed.

2 (b) [3 points] Now, suppose instead that prior to the experiment, researchers believed that $\mu = 45$ was the center of the prior distribution, with $\sigma^2 = 1$. Compute the posterior distribution for this prior (for the same data as above) for a grid 300 evenly spaced values of μ between 20 and 50 (i.e., use `seq(20, 50, length.out=300)`). Store the posterior distribution in a vector labelled, `bat_post2` and using `ggplot` reproduce the plot from above and store in as variable `bat_p2`

In [34]: Student's answer (Top)

```
n = length(x)
sigm2 = 1
nbat = mean(x)
bat_grid = seq(20, 50, length.out=300)
bat_mean2 = (nbat/sigm2 + 45/1) / (n/sigm2 + 1/1)
bat_var2 = 1 / (n/sigm2 + 1/1)
bat_prior2 = dnorm(bat_grid, 45, 1)
bat_post2 = dnorm(bat_grid, bat_mean2, bat_var2)
pp = data.frame(bat_1f=bat_grid, prior=bat_prior2, posterior=bat_post2)
df = data.frame(bat_1f=bat_grid, posterior=bat_post2)
bat_p2 = ggplot(df) +
  geom_line(aes(bat_1f=, prior, col='prior')) +
  geom_line(aes(bat_1f=, posterior, col='posterior')) +
  theme_bw()
bat_p2
# your code here
# Not yet implemented()
```

In [35]: Grade cell: c11-d6dc58c8f9f91ed Score: 6.94 / 6.94 (Top)

Hidden Tests Redacted

Congratulations! All test cases in this cell passed.

2 (i) [2 points] Multiple Choice Which response below best describe the difference between the posterior you just calculated and plotted and the one from previous parts of this problem?

Select the best response to this question and storing the number associated with the response in a variable labelled `choice`.

1. The new likelihood function has caused the center of the posterior to shift upward.
2. The posterior has caused the center of the prior to shift upward.
3. The prior has caused the center of the posterior to shift upward.

For example, if you believe the best response is #3, code: `choice=3` in the code block below.

In [36]: Student's answer (Top)

```
choice = 3
# your code here
# Not yet implemented()
```

In [37]: Grade cell: c11-25a2b302ebddbba Score: 2.78 / 2.78 (Top)

Hidden Tests Redacted

Congratulations! All test cases in this cell passed.

2 (j) [2 points] Compute a credible interval for this new posterior distribution and store it in a vector labelled, `bat_cred2`.

In [38]: Student's answer (Top)

```
bat_cred2 = c(qnorm(1-0.95/2), bat_mean2, sqrt(bat_var2)), qnorm(1-(1-0.95)/2, bat_mean2, sqrt(bat_var2)))
bat_cred2
# your code here
# Not yet implemented()
```

In [39]: Grade cell: c11-c562bfcd33da49ea Score: 2.78 / 2.78 (Top)

Hidden Tests Redacted

Congratulations! All test cases in this cell passed.

2 (k) [2 points] (True or False) We have evidence to suggest that $\mu = 30$.

1. TRUE. Given the data, and the prior, we see that $\mu = 30$ is nearly within the credible interval. It's close enough to serve as evidence to suggest $\mu = 30$.
2. FALSE. Given the data, and the prior, the probability that μ falls within $\sim (11.03, 32.57)$ is 0.05, $\mu = 30$ is outside of this interval.

If you believe this statement is true, code `response=TRUE` into the code block below. If you believe this statement to be false, code `response=FALSE` in the code block below. Be sure to use all CAPS after the = sign.

In [40]: Student's answer (Top)

```
response = FALSE
# your code here
# Not yet implemented()
```

In [41]: Grade cell: c11-3fe317a0b4c0c09f Score: 2.78 / 2.78 (Top)

Hidden Tests Redacted

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